

Today

More on Python multiprocessing, leading to image blur assignment

Kotlin / Java streams

Python currently doesn't allow more than one thread to execute code at a time

- Global Interpreter Lock (GIL)

Threading is useful in Python if threads are sleeping

Threads vs processes

- when an OS runs a program, that typically starts a new process
- a process can have many threads
- threads share memory

Processes are slower to start up, and don't share memory w/ each other

In Python (currently), we use processes to get parallelism.

Where did we use in previous assignments?

- preprocessing census data
- hash table
- STM shared resources

E.g. Census - version w/ locks

- we had one shared grid

How do you work around it?

- Communicate between processes
 - pass objects between them
 - use queues to send things back to sender
- all using messaging, which is slow

Messaging back and forth works reasonably if

- messages are few
- messages are small

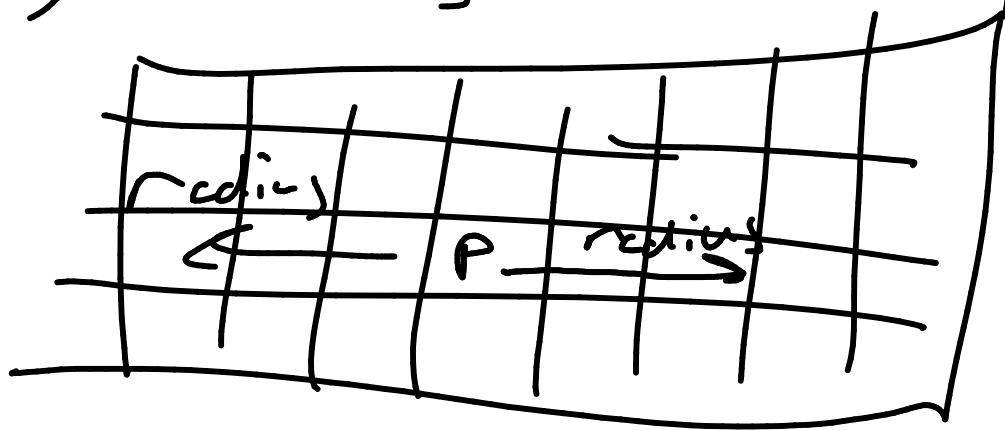
Otherwise, yuck.

Other approach is you can share memory, sort of.

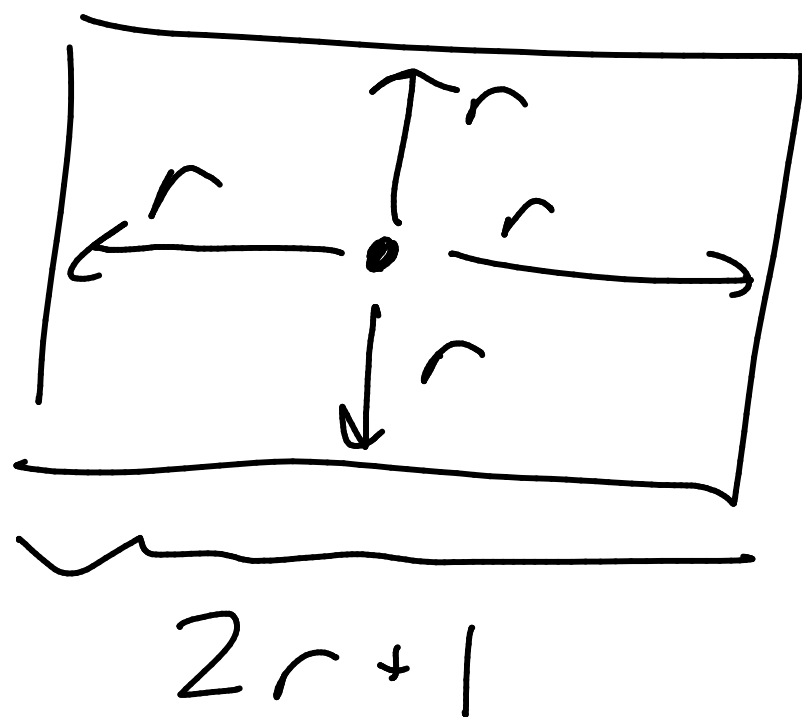
- You can create a special area of memory designated for sharing (wraps around OS tools)
- communicating w/ that shared memory is slow

Python assignment: blur an image in parallel

Blur by averaging around a pixel



make a box around pixel that goes a distance of radius in each cardinal direction



Average: add all pixel values / $(2r + 1)^2$

map works on a stream.

- it takes a function and applies it to every value in the stream.

reduce - ~~start with the first~~

